

Heart Disease Detection Using Machine Learning

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Abstract—Recent studies have highlighted a significant and concerning rise in the number of heart attacks and strokes occurring among younger populations. The long-term effect of COVID-19 on heart health is one of the factors, as patients who got infected and hospitalized due to COVID revealed a high occurrence of thrombotic events such as heart attacks and strokes. In this regard, going into the details of the necessity of early diagnosis and timely treatment in heart-related disorders, it is suggested that this can be a vital player in lowering the death rate and elevating the standard of life. To overcome this issue, we opened the Heart_Disease dataset from Kaggle and pre-processed it with a focus on handling categorical and numerical variables, dealing with the imbalance of classes, and standardizing features. We did apply numerous machine learning algorithms like Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and XGBoost. Apart from accuracy, other metrics such as precision, recall, F1-score, and confusion matrices were also used for evaluating the robustness of these models. Our research, through the construction and validation of this predictive model, demonstrates that machine learning can be a very instrumental tool in the early detection of heart disease. Such predictive models are likely to be very beneficial for healthcare workers in making choices, thereby fostering care that is proactive and, in the end, leading to deceleration of the global cardiovascular disease trend.

Keywords—Machine Learning, Logistic Regression, Heart Disease Detection, SVM, KNN, Decision Tree, Random Forest, XGBoost

I. INTRODUCTION

1.1 Background and Motivation

Heart disease is one of the major reasons for death in most parts of the world, which gradually has been influencing young people as well due to their lifestyle choices, stress, and genetics. Early and precise detection is the main point for successful treatment. However traditional diagnostic methods are often expensive, slow, and based on the interpretation of an expert. The use of healthcare data has led to a great success for machine learning (ML) which is now considered a very potent tool in unmasking hidden patterns in clinical data facilitates quick decision-making by the clinicians

1.2 Problem Statement

The problem of building accurate and generalized predictive models of heart diseases is still holding strong despite the continuous research efforts. The current studies are often plagued with numerous methodological limitations, such as improper data preprocessing, not addressing the class imbalance issue, and concentrating too narrowly on the individual algorithms. These defects also weaken the model's capacity to perform under the reality of clinical settings, thus leading to the early diagnosis of the disease to become a problem. A solid comparative framework is required to overcome these restrictions, which systematically investigates and evaluates various machine learning models to find the most efficient solution for the early and correct detection of heart disease. This will make the work of both researchers and doctors much easier as they get a dependable diagnostic and risk assessment tool.

1.3 Objectives

This research study mainly aims to carry out a detailed comparative analysis of different machine learning algorithms to evaluate their effectiveness and accuracy in predicting heart diseases. In addition, it seeks to practically examine the influence of various data preprocessing techniques on the performance, reliability, and overall dependability of the predictive models. Furthermore, the study aspires to identify and propose a trustworthy and robust model design that can simplify and enhance the accuracy of early clinical detection of heart disease, thereby contributing to better preventive healthcare outcomes

II. LITERATURE REVIEW

2.1 Previous Studies

Studies besides these have investigated utilizing machine learning alongside medical data for making predictions concerning the heart diseases.

A fresh BMC Cardiovascular Disorders paper 2025 took a machine learning approach to identifying cardiovascular risk and stressed the need for combining electronic health records (EHR) with predictive models. While pointing to the AI potential in prompt diagnosis, the work also underlined the lack of diversity in dataset samples[1].

An article in the Journal of Medical Artificial Intelligence (2024) was about the use of deep learning models in predicting cardiovascular diseases. Although quite a few such models were very accurate, the paper stated that most of them lacked the necessary clinical interpretability, which is a deciding factor in healthcare [2].

The UCI Heart Disease Dataset continues to be the leading example for using baseline data to train algorithms in heart studies. Despite its huge utility, the research shows that it reflects the patient population of those days only and, therefore, is less useful for today's population [3].

A ScienceDirect article (2025) highlighted the utilization of the hybrid machine learning method which combined decision trees with ensemble techniques that were selected for use in cardiovascular prediction. The research results were highly effective; however, they experienced difficulties in on-the-ground implementation and external validation [4].

A novel Nature Scientific Reports study (2025) featured explainable AI models that could accurately detect cardiovascular diseases. The intervention of this research was in the improvement of the interpretability; however, it still disclosed a computational challenge of the scalability of big, real-world datasets [5].

The study on PubMed Central (2024) delved into the influence that diseases, including those that have Covid-19 as their cause, have on the cardiovascular system. The study found that post-COVID-19 patients suffered from heart attacks and strokes at almost twice the rate, hence, it is vital to introduce those emerging risk factors into the prediction models [6].

Statements by the American Heart Association (2024) proclaimed that infection with COVID-19 is a significant cause of the long-term risk for cardiovascular diseases. The evidence points to the fact that current prediction models need to be adapted to consider the post-pandemic health dynamics [7].

2.2 Strengths and Gaps in Previous Studies

Present research illustrates the efficiency of machine learning in predicting heart disease, especially when using well-organized data sets and ensemble techniques. But along with the benefits, the research also exhibits the limitations that are as follows:

- Dataset limitations – The majority of models constructed depend on UCI or similar datasets, which are outdated and inadequate in terms of present-day population reflection.
- Interpretability vs. accuracy trade-off – The models holding top performance have the clinical interpretability problem, thus their integration into the healthcare sector is complicated.

- Integration of emerging factors – There are many models in which newly appearing risk factors like post-COVID cardiovascular complications are not considered.

2.3 Our Approach

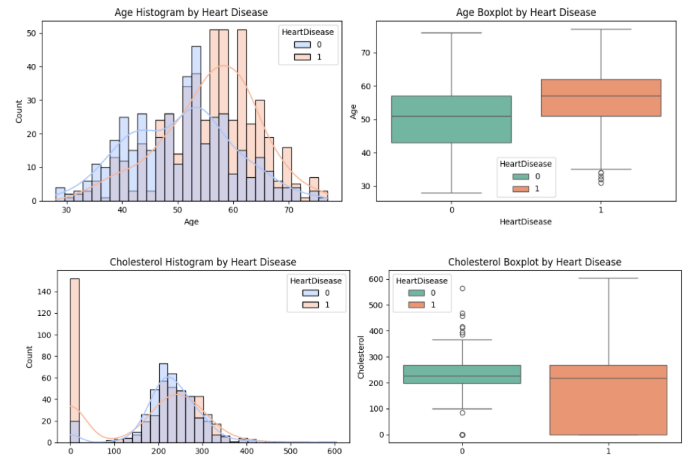
In order to find solutions to those problems, we have carried out a study that uses a balanced dataset with up-to-date risk variables and that designs machine learning models to be interpretative-oriented. Furthermore, we put first the inclusion of both traditional and new health indicators to make the predictions not only adequate but also useful in clinical practice.

III. III. METHODOLOGY

3.1 Data Preparation and Preprocessing

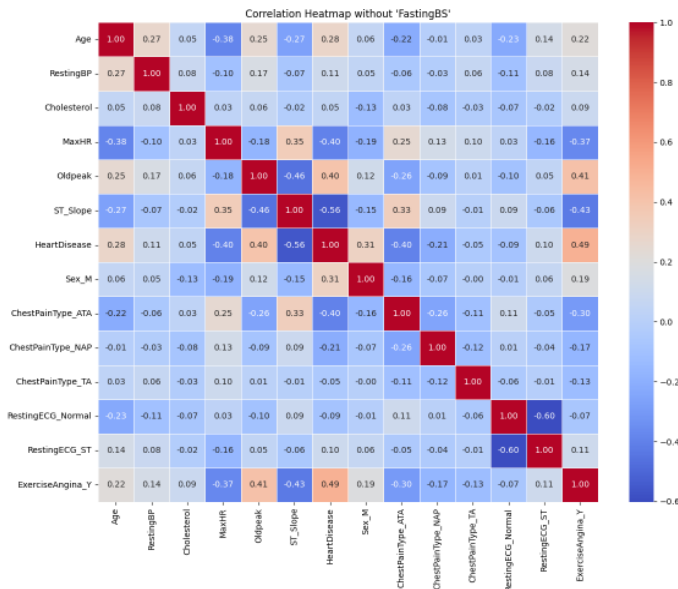
The Heart_Disease dataset from Kaggle consists of both descriptive and numerical data referring to the human cardiovascular system. The data has been thoroughly processed to ensure quality and consistency.

- The very first point in the data processing was the finding of missing and erroneous values in the data through the cleaning process. Variables were divided into numerical and categorical groups to make the next processing easier.
- The **SMOTE (Synthetic Minority Over-sampling Technique)** method was implemented to handle the class imbalance issue.
- Next, StandardScaler was applied to each feature to normalize all features to the same scale.



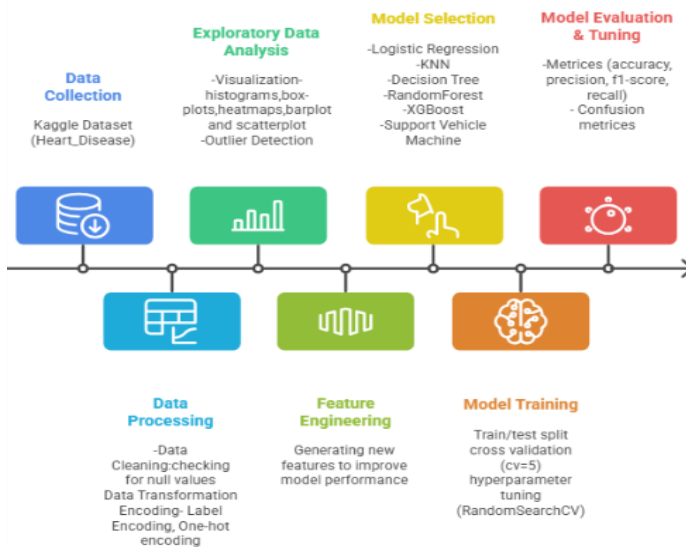
Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) was implemented using statistical and pictorial instruments to know distributions and get a general idea of the data.
- Besides that, the bar graphs, scatter plots, histograms, and pie charts were used. Outliers are detected and converted into median and mean.



Feature Engineering

- One-Hot Encoding was the way used to change categorical features to a binary numerical format.
- Feature selection was accomplished by removing those features that had the lowest correlation with the target variable, as these had little impact on the model's predictive power.



A.

3.2 Model Building

- Several machine learning techniques were tried, and the results were compared. These models were Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree Classifier, Random Forest Classifier, and XGBoost Classifier. Hyperparameter tuning was conducted using RandomizedSearchCV to get the best possible model.

- Of all the models, the XGBoost Classifier was the most successful; it managed to reach the goal of 86.4% accuracy and gave precision of 91%. This makes XGBoost a particularly valuable tool in health care prediction projects, since it lowers the number of false positive results, i.e., the number of unnecessary medical procedures will be reduced.

3.3 Evaluation Metrics

One of the ways to check how well a model would work is to evaluate it on different metrics namely: Accuracy, Precision, Recall and F1-Score. The selection of these metrics caters to the need for diversified m classification based on imbalanced data sets.

Table1: Logistic Regression results

Classification Report:				
	precision	recall	f1-score	support
0	0.83	0.87	0.85	82
1	0.89	0.85	0.87	102
accuracy			0.86	184
macro avg	0.86	0.86	0.86	184
weighted avg	0.86	0.86	0.86	184
Accuracy Score: 0.8586956521739131				

Table 2: KNN results

Classification Report:				
	precision	recall	f1-score	support
0	0.84	0.84	0.84	82
1	0.87	0.87	0.87	102
accuracy			0.86	184
macro avg	0.86	0.86	0.86	184
weighted avg	0.86	0.86	0.86	184
Accuracy Score: 0.8586956521739131				

Table 3: SVM results

Classification Report:				
	precision	recall	f1-score	support
0	0.83	0.85	0.84	82
1	0.88	0.86	0.87	102
accuracy			0.86	184
macro avg	0.86	0.86	0.86	184
weighted avg	0.86	0.86	0.86	184
Accuracy Score: 0.8586956521739131				

Table 4: Decision Tree results

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Classification Report:
              precision    recall  f1-score   support

     0       0.75       0.77       0.76         82
     1       0.81       0.79       0.80        102

 accuracy          0.78          0.78          0.78        184
 macro avg          0.78          0.78          0.78        184
 weighted avg          0.78          0.78          0.78        184

Accuracy Score: 0.782608695652174

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Table 5: Random Forest results

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Classification Report:
              precision    recall  f1-score   support

     0       0.82       0.89       0.85         82
     1       0.91       0.84       0.87        102

 accuracy          0.86          0.86          0.86        184
 macro avg          0.86          0.87          0.86        184
 weighted avg          0.87          0.86          0.86        184

Accuracy Score: 0.8641304347826086

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Table 6: XGBoost results

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Classification Report:
              precision    recall  f1-score   support

     0       0.82       0.89       0.85         82
     1       0.91       0.84       0.87        102

 accuracy          0.86          0.87          0.86        184
 macro avg          0.86          0.87          0.86        184
 weighted avg          0.87          0.86          0.86        184

Accuracy Score: 0.8641304347826086

```

3.4 Experimental Setup

All the experiments were conducted using Python in Google Collab. The main libraries were Pandas and NumPy for data manipulation, Matplotlib and Seaborn for visualization, Scikit-learn for preprocessing, model training, and evaluation, XGBoost for gradient boosting, and Imbalanced-learn for class balancing via SMOTE.

The computational environment was made up of Google Collab's cloud-based GPU support with standard runtime configurations. The division of data into training and testing sets was done in an 80:20 ratio. During training, 5-fold cross-validation was applied to check model generalizability.

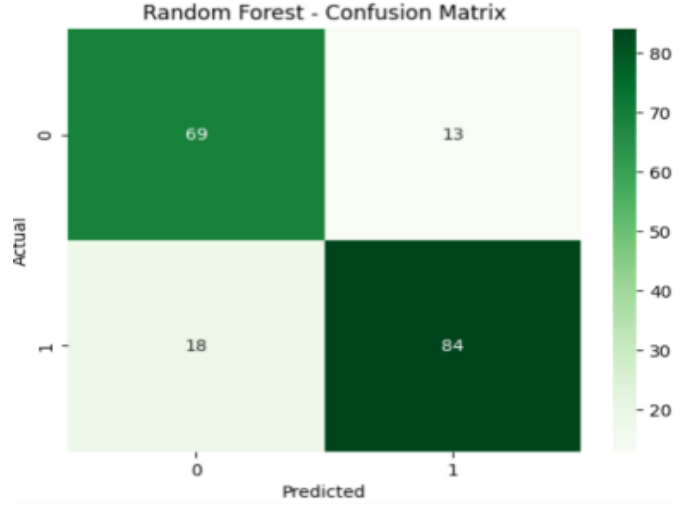
IV.

V. IV. RESULTS

After the models were created, their accuracies were evaluated: Logistic Regression – 85.86%, KNN – 85.86%, SVM – 85.87%, Decision Tree – 78.26%, Random Forest – 83.15%, and XGBoost – 86.41%. The comparative analysis revealed that ensemble-based methods like Random Forest and XGBoost

provided superior predictive performance compared to traditional models, though simpler models like Logistic Regression offered strong interpretability.

Confusion Matrix 1: Random Forest



Confusion Matrix 2: XGBoost

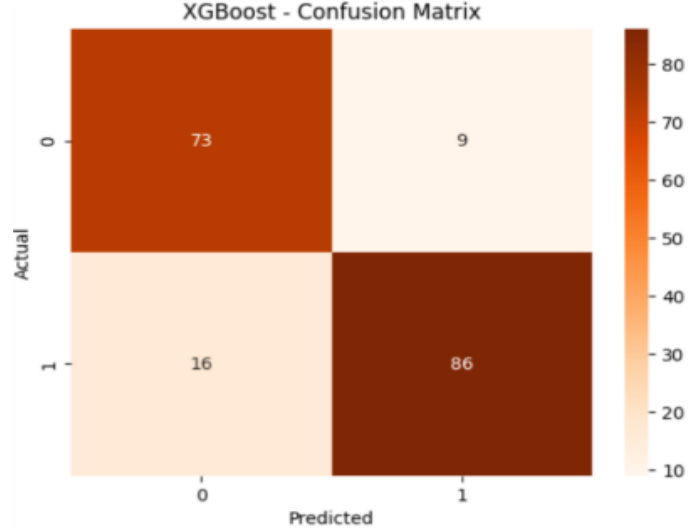


TABLE I. CONFUSION MATRIX RESULTS

MODELS	TRUE POSITIVE	TRUE NEGATIVE	FALSE POSITIVE	FALSE NEGATIVE
LOGISTIC REGRESSION	83	71	11	19
K- NEAREST NEIGHBOURS	87	72	10	15

SUPPORT VECTOR MACHINE	86	71	11	16
DECISION TREE	85	63	19	17
RANDOM FOREST	86	71	11	16
XG BOOST	88	73	9	14

V. DISCUSSION

Most of the models, including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), that were evaluated in this research, have achieved very close accuracies as their main result, i.e. from 85% to 86% of about. XGBoost has better performance, with a maximum accuracy of 86.41%, being the only model to stand out significantly. This suggests that both kinds of models, those that are simple and those that are highly complex, can have success in heart disease prediction.

One of the exceptions in this research is the Decision Tree model, which was considerably weaker and had a performance of 78.26% accuracy. Such an outcome tends to speak of single, non-ensemble models that may not generalize as well when confronted with the inherent complexities of the dataset. Besides, the performance of Logistic Regression, which was only slightly behind more advanced models, indicates that simpler, more interpretable models have the potential to achieve substantial accuracy. This is of an utmost importance since these models are interpretable just like in healthcare and medical diagnosis, where the reason behind a prediction can be as significant as the prediction itself.

Despite the encouraging outcomes, the current research has several limitations. The use of accuracy as the only metric for evaluation of performance is insufficient for dealing with critical diagnostic scenarios. In the medical field, metrics such as precision, recall, and the F1-score are very important for understanding a model's performance on both positive and negative cases. Although such obstacles have been encountered,

the current findings are a strong indication of the machine learning potential for early prediction and clinical decision-making. Subsequent studies should focus on achieving a balance between predictive power and model interpretability to facilitate practical application in real-world scenarios.

VI. CONCLUSION AND FUTURE WORK

This research made use of the Heart_Disease dataset, which had undergone preprocessing with correlation analysis, data encoding, outlier detection, and replacing extreme values with the mean. The findings revealed that among all the models that were put into practice, XGBoost was the one that scored the highest accuracy of 86.4%.

However, these results still leave some room for improvement. The model's efficiency can be refined by deploying a larger as well as a more varied dataset to boost its generalizability. Thus, the future work should also focus on the implementation of new methods like advanced feature engineering and deep learning for getting closer predictions to the true values. Also, the employment of explainable AI tools might provide a way into understanding the exactness of the predictions of the doctors, and in this sense, it would be a step closer to trustworthiness - a very important feature in the medical field.

VII. REFERENCES

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